CSE4144 Information Retrieval · Project 13

Legal Retrieval using CaseHOLD

Comparing TF-IDF, BM25, a Sentence Transformer and Hybrid RRF for finding the holding a court cites

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The problem: finding the holding a court relies on

Courts back every legal claim with a citation to an earlier case.

The citation is often followed by its holding: what that case actually decided.

A lawyer needs the reverse: given the argument, find the case whose holding supports it.

Our task: query = the citing passage, documents = holding statements, goal = rank the cited holding first.

Query (part of the citing passage)

"…the anticipation of increased interference or potential hazards on the land remaining after a taking, which one court has labeled 'stigma damages,' can be considered in assessing severance damages. See, e.g., Vector Pipeline v. 68.55 Acres of Land (<HOLDING>)"

Document we should return

holding that stigma damages were properly included in calculation of severance damages where proximity of gas pipeline could trigger fears about possible mishaps

Dataset: CaseHOLD

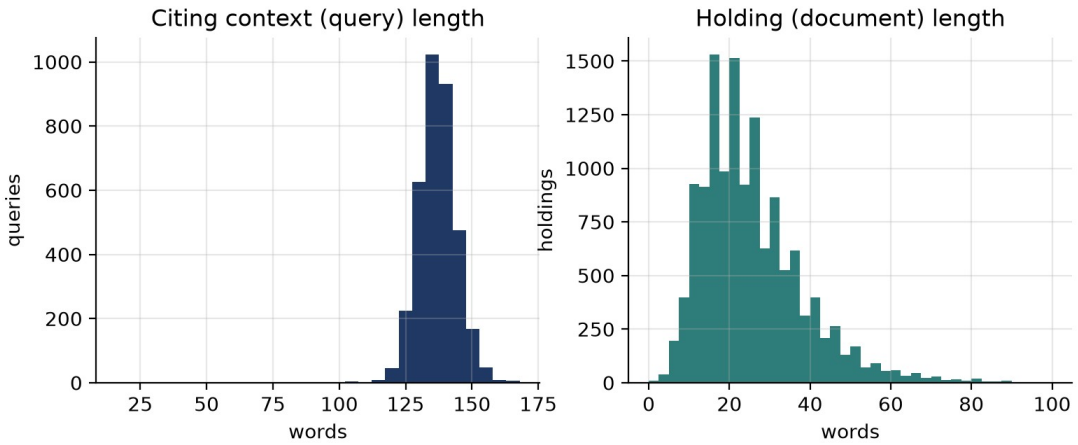
Split	Queries	Unique holdings	Words / query	Words / holding
Train	45,000	50,291	136.3	26.4
Validation	3,900	14,138	136.3	25.4
Test	3,600	13,297	136.4	25.4

From US federal and state court decisions (Harvard Law Library corpus).

Each question: citing passage + 5 holdings, exactly one correct.

The 4 wrong holdings were picked by TF-IDF similarity to the right one, to make the task hard.

We use the LexGLUE copy (coastalcph/lex_glue, case_hold).



Word counts in the test split: long queries, short holdings

One test question, as the models see it

Citing passage (end): …have held that the anticipation of increased interference or potential hazards on the land remaining after a taking, which one court has labeled “stigma damages,” can be considered in assessing severance damages. See, e.g., Vector Pipeline, L.P., v. 68.55 Acres of Land, 157 F.Supp.2d 949, 957 (N.D.Ill.2001) (<HOLDING>)

0	holding that actual damages under respa included emotional damages	
1	holding that increased fear of flooding was a matter that factfinder could properly consider in assessing severance damages	
2	holding that local buyers fear of possible knapweed infestation legitimately affected calculation of severance damages	
3	recognizing concept of stigma damages as they apply to property which is in fact contaminated	
4	**holding that stigma damages were properly included in calculation of severance damages where proximity of gas pipeline could trigger fears about possible mishaps**	**correct**

Turning a multiple-choice dataset into a search task

Full-pool setting (main)

Documents: all 13,297 unique holdings of the test split

Queries: 3,600 citing passages

Exactly 1 relevant holding per query

Metrics: Precision@10, Recall@k, MAP, nDCG@10

5-choice setting (original task)

Each query ranks only its own 5 options

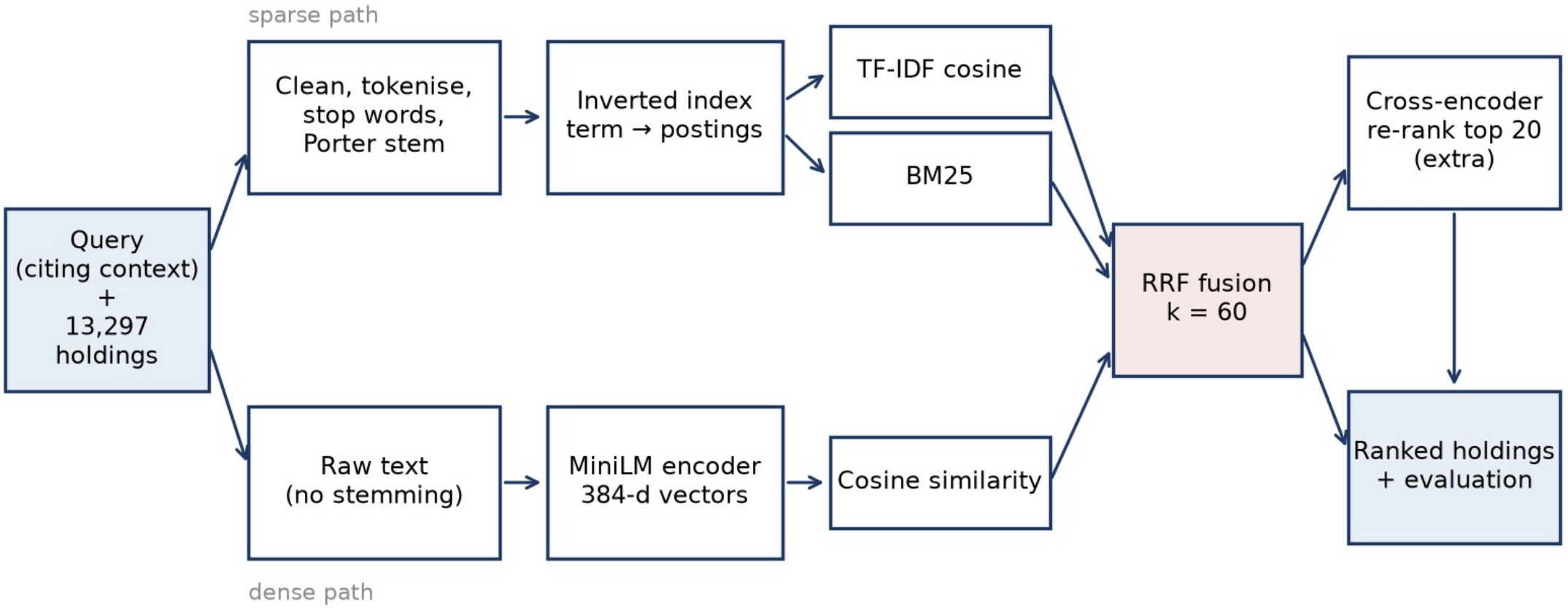
Accuracy = correct option ranked first

Random guessing = 0.20

Hard: options were chosen to look alike

Protocol: every parameter was tuned on the 3,900 validation queries; the test split was used once, at the end. The train split was used only for the fine-tuning extension.

System pipeline



Preprocessing and the inverted index

1. Lowercase, remove <HOLDING>
2. Tokenise; drop tokens with digits (citations)
3. Remove stop words + legal fillers (holding, v, see, id…)
4. Porter stemming

Holding: “holding that stigma damages were properly included in calculation of severance damages”

Tokens: stigma, damag, properli, includ, calcul, sever, damag

Inverted index on the test pool

	Value
Holdings indexed	13,297
Vocabulary (stemmed terms)	7,233
Postings (term, holding) pairs	150,284
Avg. terms per holding	12.56
Avg. postings list length	20.78

Validation check (TF-IDF MRR)

no stop words, no stemming: 0.283

stop words only: 0.279

stop words + stemming: 0.306

The four ranking models

TF-IDF

Vector of (1 + log tf) × idf weights, ranked by cosine similarity.

BM25

Probabilistic model with term saturation (k1) and length normalisation (b). Our own implementation on the inverted index.

Sentence Transformer

all-MiniLM-L6-v2: text → 384-d vector; cosine similarity. Captures meaning, not exact words.

Hybrid RRF

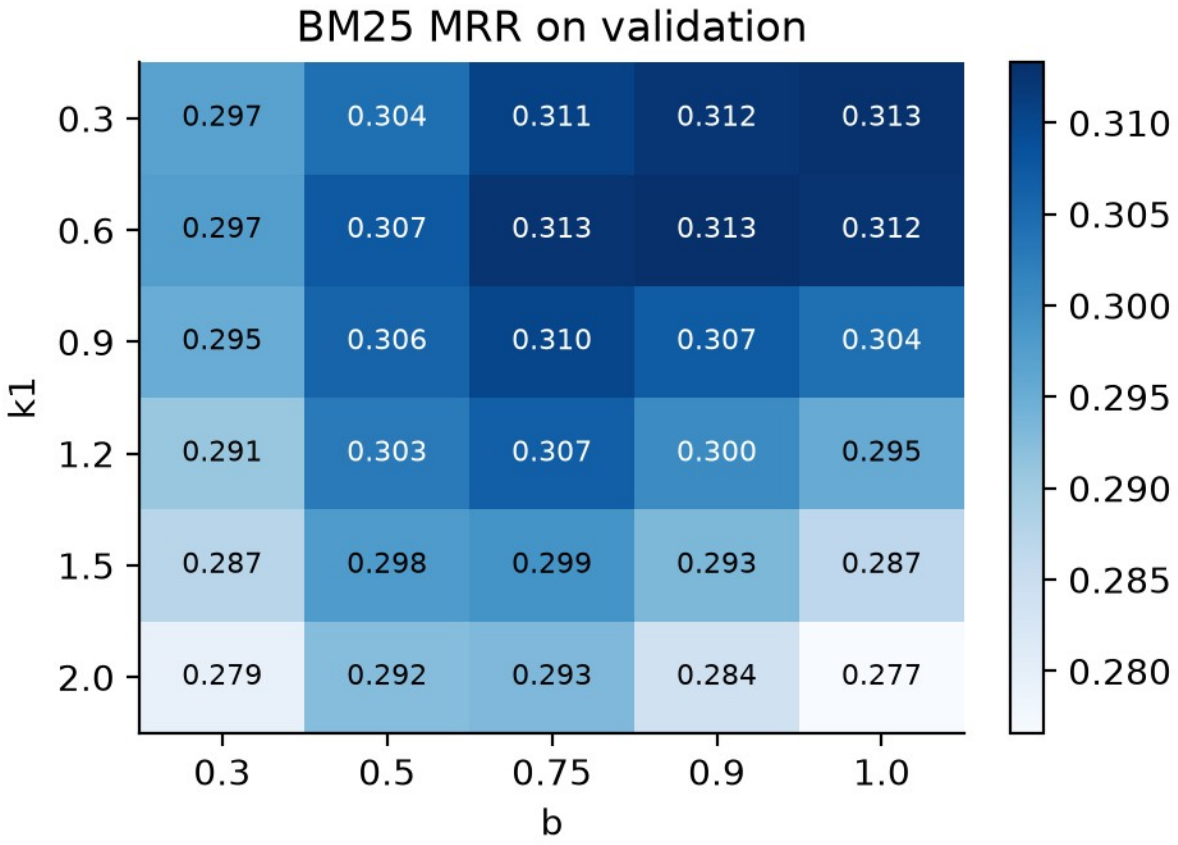
Merges the top-1000 lists of the three models by rank only, so no score normalisation is needed.

$$\text{BM25}(q, d) = \Sigma\ \text{idf}(t) \cdot \text{tf}\cdot(k1 + 1) \,/\, (\ \text{tf} + k1\cdot(1 - b + b\cdot|d|/\text{avgdl})\ )$$

$$\text{RRF}(d) = \Sigma \text{ over models } m \text{ of } \ 1 \,/\, (k + \text{rank}_m(d)), \quad k = 60$$

Metrics: one relevant holding per query, so MAP = MRR and the best possible Precision@10 is 0.1. nDCG@10 = 1 / log2(rank + 1) inside the top 10.

Choosing settings on the validation split



Query: whole passage (MRR 0.307) beat last 64 words (0.290) and last 32 (0.214)

BM25: k1 = 0.6, b = 0.9 (short documents: low k1, strong length normalisation)

Dense model: MiniLM 0.234 vs BGE-small 0.182

RRF k: k = 1 → 0.331, k = 60 → 0.317; we report both

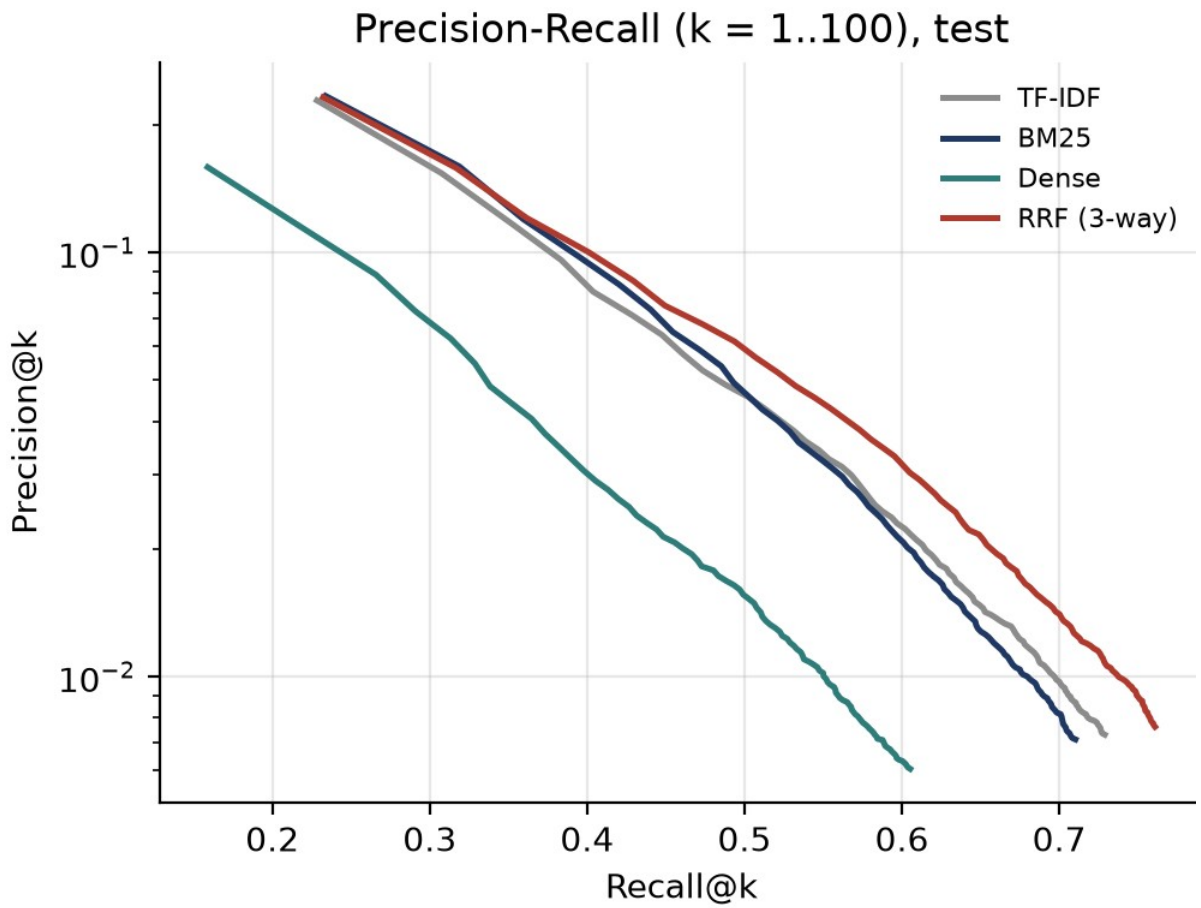
Results: full-pool search (3,600 queries, 13,297 holdings)

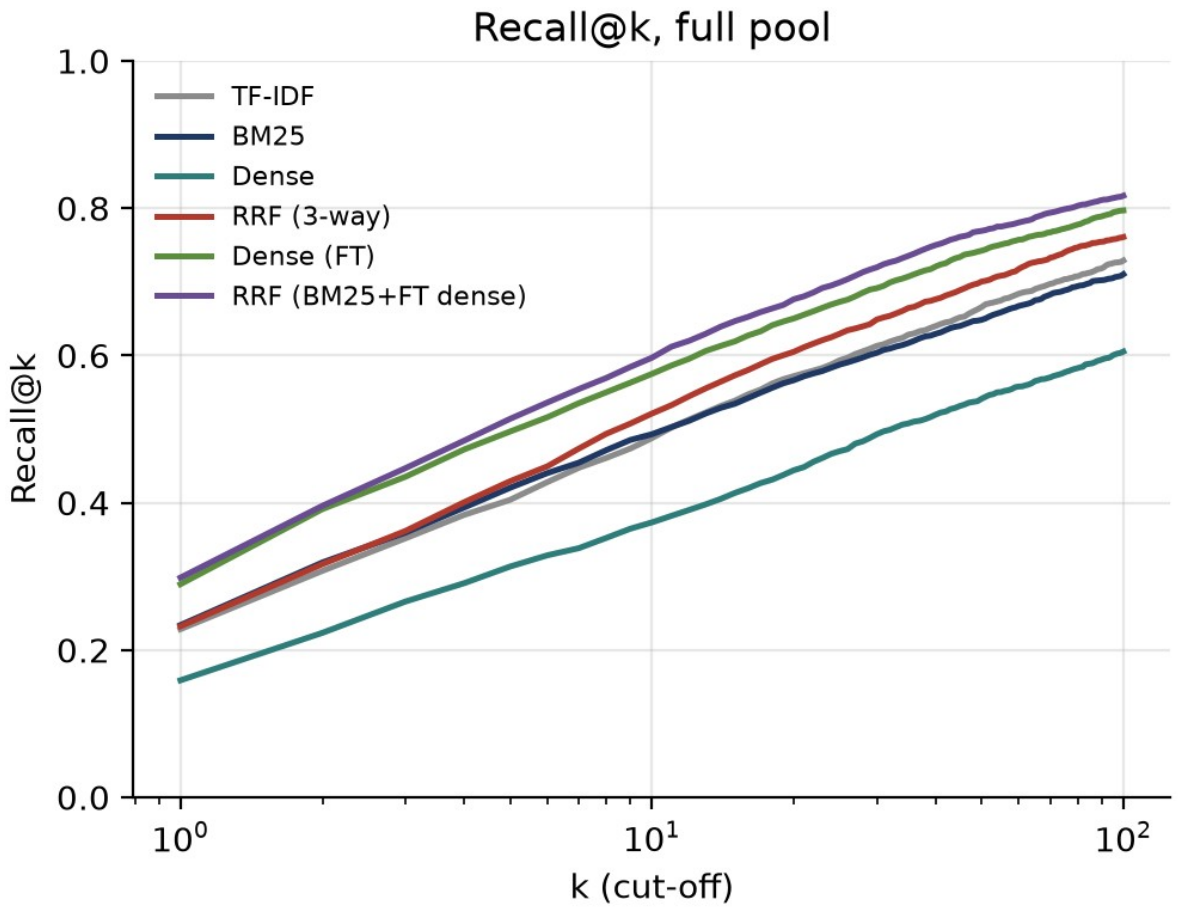
Ranking model	Precision@10	Recall@10	Recall@100	MAP (= MRR)	nDCG@10
TF-IDF	0.049	0.488	0.729	0.317	0.349
BM25	0.049	0.493	0.710	0.323	0.356
Sentence Transformer	0.037	0.373	0.605	0.234	0.260
Hybrid RRF (k = 60, standard)	0.052	0.521	0.761	0.328	0.365
Hybrid RRF (k = 1, tuned on validation)	**0.053**	**0.530**	**0.759**	**0.342**	**0.379**

BM25 is the strongest single model; the off-the-shelf dense model is clearly weaker on legal text.

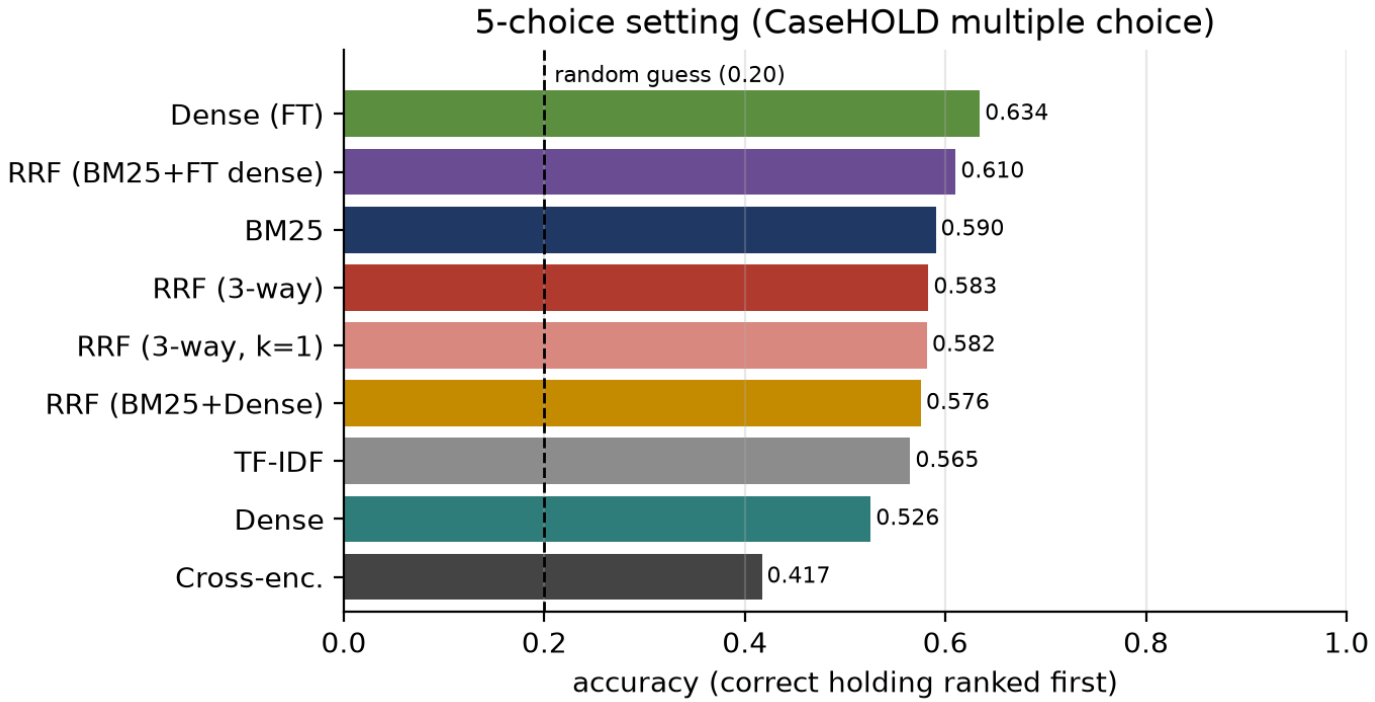
The hybrid mainly improves recall: Recall@100 0.710 → 0.761, MAP 0.323 → 0.328 (Wilcoxon $p < 0.001$ vs BM25).

Precision-Recall graph and Recall@k





Results: the original 5-choice task



Random guess = 0.20

BM25 0.590 beats the 3-way hybrid 0.583 here.

Reason: the wrong options were chosen by TF-IDF similarity, so all five look alike to word matching.

Best: fine-tuned dense model, 0.634.

Published reference: BiLSTM about 0.4 and BERT about 0.6 macro F1 (Zheng et al., different split and metric).

Extensions: fine-tuning helps, a web re-ranker hurts

Run (full pool, test)	Recall@10	Recall@100	MAP	nDCG@10
BM25 (reference)	0.493	0.710	0.323	0.356
Hybrid RRF (TF-IDF + BM25 + Dense)	0.521	0.761	0.328	0.365
Hybrid + ms-marco cross-encoder	0.407	0.761	0.196	0.231
Sentence Transformer (base)	0.373	0.605	0.234	0.260
Sentence Transformer, fine-tuned	0.575	0.797	0.389	0.426
Hybrid RRF (BM25 + fine-tuned)	**0.597**	**0.817**	**0.400**	**0.440**

Fine-tuning

45,000 train pairs, 1 hard negative each, multiple negatives ranking loss, 1 epoch, ~11 min.

Cross-encoder

ms-marco-MiniLM re-scores the top 20. Trained on web search, so it does not transfer to legal text.

Takeaway

Domain data mattered more than a bigger or fancier model.

Error analysis: where and why the models fail

474 queries where only BM25 ranks the answer first

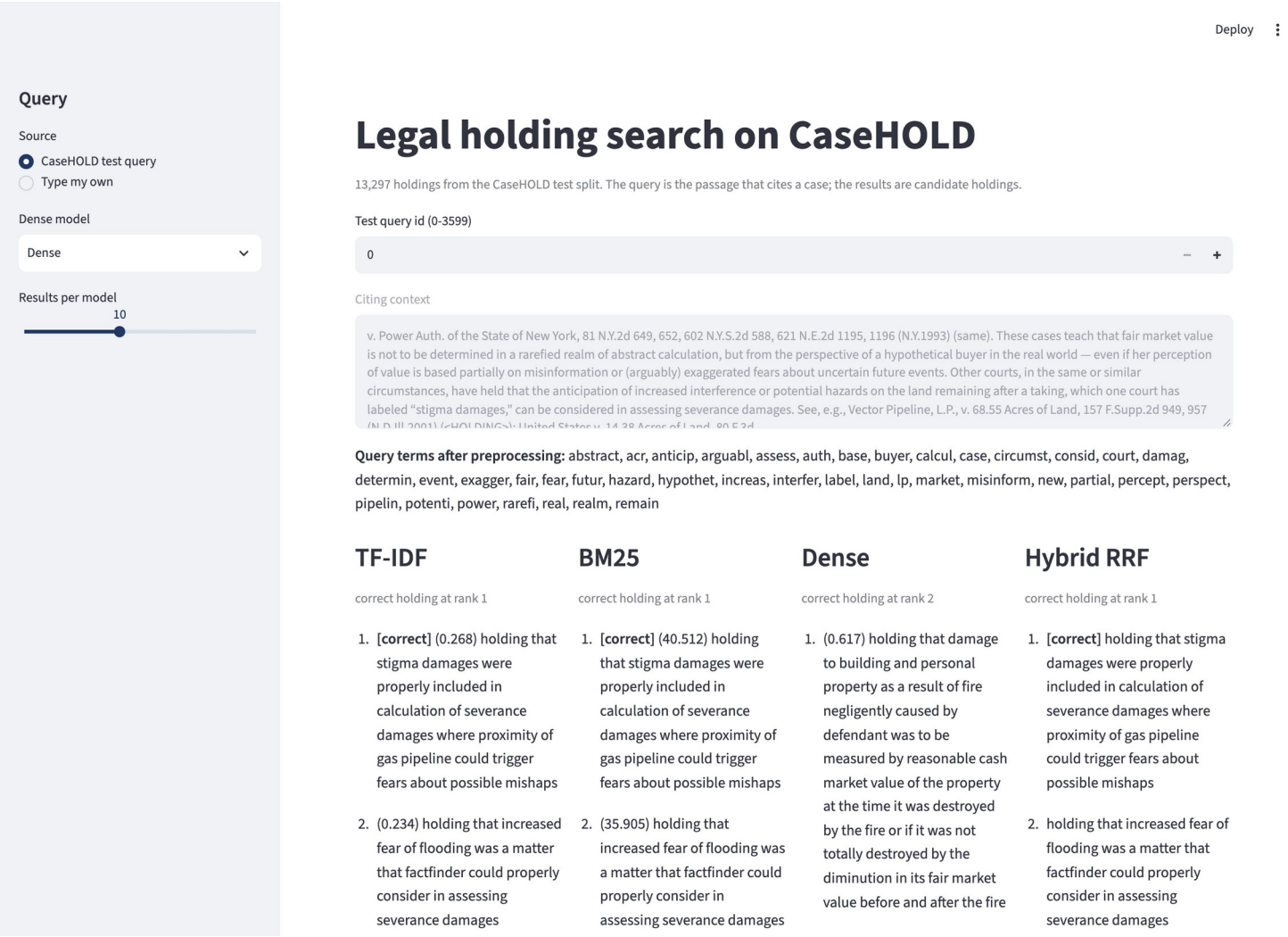
204 queries where only the dense model does

44% of queries: neither model has it in the top 10

Query	BM25 rank	Dense rank	What happened
3185	1	129	Rare word “structuring” matched exactly
296	1,694	1	Same meaning, no shared keywords
2885	3,475	6,823	Passage is about a strip search; the cited holding is about court procedure

Longer holdings are easier: BM25 MRR 0.213 for holdings of 15 words or less vs 0.402 above 40 words.

Search interface (live demo)



streamlit run app/search_app.py

Pick a test query or type any passage.

Shows the stemmed query terms.

Top results of TF-IDF, BM25, dense and hybrid side by side.

Marks the correct holding when it is known.

Base or fine-tuned dense model from the sidebar.

Conclusion and limitations

What we found

BM25 (MAP 0.323) beat the off-the-shelf dense model (0.234).

RRF mainly raised recall: Recall@100 0.710 → 0.761.

Fine-tuning on CaseHOLD: dense 0.234 → 0.389; with BM25 0.400, our best.

A web-trained cross-encoder made results worse.

Limitations

One relevant holding per query; Precision@10 capped at 0.1.

Pool = test-split holdings, not a full case law database.

Small models, 1 training epoch, one laptop.

Next steps

Legal-domain encoder (e.g. Legal-BERT).

Train the re-ranker on CaseHOLD itself.

Weighted RRF instead of equal weights.

References

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